GS-CRC: Generalised Symmetric Cluster-Repelling Contrastive
Loss for Robust Online Intrusion Detection

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Abstract*—We identify seven structural pathologies in the AOC-IDS framework (Zhang et al., INFOCOM 2024): five in the Cluster-Repelling Contrastive (CRC) loss and two in the detection pipeline, and prove that each degrades performance in a quantifiable, theoretically predicted way. The central loss defects are: (i) exponential minority-subclass gradient suppression at low temperature τ, verified in closed form; (ii) Fisher inconsistency—the CRC population minimiser does not constrain abnormal-class geometry, so useful sub-structure depends entirely on the reconstruction term and collapses when that term dominates; (iii) equal encoder/decoder weighting with no principled justification. The pipeline defects are: (iv) fitting a single Gaussian to a multi-modal abnormal score distribution incurs $\Omega(1)$ excess Bayes risk regardless of sample size; (v) the static normal centroid cannot track benign traffic drift. We propose the *Generalised Symmetric Cluster-Repelling Contrastive* (GS-CRC) loss with four fixes to the CRC objective, and augment the detection pipeline with a K-component GMM discriminator (BIC-selected), an EMA normal centroid, and a KL-divergence-triggered online update cadence. We further replace majority-vote decision with calibrated log-likelihood-ratio (LLR) fusion and reduce the autoencoder bottleneck to the information-theoretically motivated dimension $m^* = \lceil d^* + \log_2 K \rceil$. On UNSW-NB15 at 100 training epochs (one-eighth of the paper's standard 800), our full system raises mean F1 from 0.301 to 0.868 (+188.2%), recall from 0.380 to 0.954, eliminates all five degenerate seeds that AOC-IDS produces, and reduces F1 standard deviation from 0.315 to 0.003. GS-CRC strictly recovers CRC at $\gamma = 0$, $\alpha_{\mathrm{symm}} = 0$, $\rho = 1$, constant τ, and $\mathcal{L}_{\max} = \infty$.

***Index Terms*—intrusion detection, contrastive learning, online learning, imbalanced classification, Fisher consistency, focal modulation, Gaussian mixture model, UNSW-NB15.**

I. Introduction

Online network intrusion detection demands a classifier that learns continuously from unlabelled traffic, propagating pseudo-labels into its own training set. AOC-IDS [1] is the state-of-the-art framework for this setting: an autoencoder trained with the Cluster-Repelling Contrastive (CRC) loss maps flows into a space where a Gaussian mixture discriminator operates without requiring per-flow labels at deployment.

AOC-IDS reports strong aggregate numbers on NSL-KDD and UNSW-NB15, yet our replication reveals three operationally significant failure modes. First, on UNSW-NB15 at the authors' own hyperparameters, 3 of 5 random seeds collapse to a degenerate classifier—one that predicts every sample as a single class. Second, even non-degenerate seeds show extreme instability (F1 standard deviation = 0.315). Third, the implementation is incompatible with current GPU hardware without patching four separate runtime bugs.

We trace each failure to a provable structural property of the CRC loss or the Gaussian discriminator, and fix each with a minimal, theoretically motivated intervention. The result is GS-CRC plus an upgraded detection pipeline that together address seven pathologies.

Contributions.

- A gradient analysis of CRC (Lemma 1 and Theorem 2) showing exponential per-subclass gradient suppression and proving CRC is Fisher-inconsistent (Theorem 3).
- The GS-CRC loss (Definition 1) with proof of Fisher consistency (Theorem 5) and polynomial gradient balance (Theorem 6), plus an adaptive temperature schedule (Theorem 4).
- A formal excess-risk bound for the 2-component Gaussian discriminator (Theorem 7), motivating the K-component GMM fix, and a calibrated LLR decision rule (Proposition 8).
- Systems-level improvements with formal guarantees: a KL-divergence drift trigger (Theorem 10) and an EMA normal centroid with a drift-tracking error bound (Theorem 11).
- Empirical validation: F1 $0.301 \rightarrow 0.868$ (+188.2%), recall $0.380 \rightarrow 0.954$, zero degenerate seeds, F1 std $0.315 \rightarrow 0.003$.

II. Related Work

Contrastive learning for NIDS. AOC-IDS [1] is the first work to apply an online contrastive objective to network traffic. Earlier NIDS systems rely on reconstruction loss alone [3] or supervised cross-entropy on labelled datasets [4].

Contrastive loss analysis. Wang and Liu [5] analyse supervised contrastive loss in the balanced regime. Our minority gradient starvation analysis is specific to the extreme imbalance of NIDS ($r = l_a/l_n \ll 1$) and has no direct precedent.

Fisher consistency. Bartlett and Tewari [6] formalise Fisher consistency for margin losses. We apply the definition to the contrastive setting where the "classifier" is a downstream Gaussian mixture.

Bounded influence and noise robustness. Huber [7] defines B-robustness through a bounded influence function. We apply this to the pseudo-label feedback loop, connecting it to Natarajan et al.'s [8] noise-tolerance bounds.

Uncertainty-weighted multitask. Kendall et al. [9] derive the principled weighting $1/(2\sigma^2)$ for homoscedastic multitask losses. We apply this to the encoder/decoder loss combination of AOC-IDS.

III. CRC Loss: Analysis and Pathologies

A. Notation

Let $\mathcal{X} \subset \mathbb{R}^d$ be the input space. Let $f_\theta : \mathcal{X} \to \mathbb{R}^m$ be the encoder with unit-normalised output $z_i = f_\theta(x_i)/\|f_\theta(x_i)\|$, so $h(i,j) = z_i^\top z_j \in [-1,1]$. Labels $y_i \in \{0,1\}$ with 0 = normal, 1 = abnormal; l_n, l_a are empirical batch counts; $r = l_a/l_n$. The CRC loss [1] for a mini-batch $\mathcal{B}$:

$$\mathcal{L}_{ij}^{\text{CRC}} = -\log \frac{e^{h(i,j)/\tau}}{e^{h(i,j)/\tau} + S}, \quad S = \sum_{p=1}^{l_n} \sum_{k=1}^{l_a} e^{h(p,k)/\tau}, \quad (1)$$

$$\mathcal{L}^{\text{CRC}} = \frac{1}{l_n(l_n-1)} \sum_{i \in \mathcal{B}_n} \sum_{j \neq i, j \in \mathcal{B}_n} \mathcal{L}_{ij}^{\text{CRC}},$$

where $\tau = 0.02$ and summation runs over normal-anchor, normal-positive pairs.

B. Gradient Analysis

Lemma 1 (CRC per-sample gradients)**.** *For pair (i,j) with $Z_{ij} = e^{h(i,j)/\tau} + S$, define $p_{ij} = e^{h(i,j)/\tau}/Z_{ij}$ and $q_{pk}^{(ij)} = e^{h(p,k)/\tau}/Z_{ij}$. Then:*

$$\frac{\partial \mathcal{L}_{ij}}{\partial z_i} = \frac{1}{\tau} \left[(p_{ij} - 1)\, z_j + \sum_{k=1}^{l_a} q_{ik}^{(ij)}\, z_k \right], \quad (2)$$

$$\frac{\partial \mathcal{L}_{ij}}{\partial z_m} = \frac{1}{\tau} \sum_{p=1}^{l_n} q_{pm}^{(ij)}\, z_p \qquad (y_m = 1). \quad (3)$$

Aggregating over all anchor pairs:

$$\frac{\partial \mathcal{L}^{CRC}}{\partial z_m} = \frac{\bar{Q}}{\tau} \sum_{p=1}^{l_n} e^{h(p,m)/\tau}\, z_p, \quad (4)$$

where $\bar{Q} = \frac{1}{l_n(l_n-1)} \sum_{(i,j)} Z_{ij}^{-1}$.

Proof. Differentiate $-h_{ij}/\tau + \log Z_{ij}$ using $\nabla_{z_a} h(a,b) = z_b$ (since $\|z\| = 1$). For the abnormal gradient, z_m appears only inside S; collecting terms yields (3). Aggregating over (i,j) pairs gives (4). □

Equation (3) reveals the first structural defect: *the abnormal gradient is purely repulsive*. There is no analogue of the cohesive term in (2). Whatever structure exists within the abnormal class is not regularised by CRC; it depends entirely on the decoder reconstruction term.

C. Pathology 1: Exponential Minority Gradient Starvation

Theorem 2 (Subclass gradient suppression)**.** *Partition abnormals into subclasses $\{c\}$ with sizes $l_a^{(c)}$ and mean cosine similarity to normals $\bar{h}^{(c)}$. Let $h^{(\max)} = \max_c \bar{h}^{(c)}$. In the asymptotic regime where the dominant subclass governs S:*

$$\left\| \frac{\partial \mathcal{L}^{CRC}}{\partial z_m} \right\| \le \frac{e^{(\bar{h}^{(c)} - h^{(\max)})/\tau}}{\tau\, l_a^{(\max)}} \cdot (1 + o(1)). \quad (5)$$

Proof. $S \approx l_n l_a^{(\max)} e^{h^{(\max)}/\tau}$ by the Laplace approximation as $\tau \to 0$, giving $\bar{Q} \approx S^{-1}$. Substituting into (4) and bounding the numerator by $l_n e^{\bar{h}^{(c)}/\tau}$ yields (5). □ □

Remark 1. *At $\tau = 0.02$ and gap $\Delta = 0.10$ (conservative for UNSW-NB15 low-frequency subclasses), suppression factor $e^{-5} \approx 6.7 \times 10^{-3}$: minority gradients are attenuated two orders of magnitude* per sample*, exponentially in $1/\tau$.*

D. Pathology 2: Fisher Inconsistency

Theorem 3 (CRC is Fisher-inconsistent)**.** *Under any joint distribution P with $\pi_a > 0$, the population risk $\mathcal{L}^{CRC,pop}$ is minimised by the* degenerate *family mapping all normals to $z_n^\star$ and all abnormals to $-z_n^\star$. Among non-degenerate solutions, CRC has* no preference *between abnormals forming a single blob versus stratifying by subclass.*

Proof. Substituting $h(i,j) = 1$ for positive pairs and $h(p,k) = -1$ for negative pairs: $\mathcal{L}_{ij}^{\text{CRC}} = \log(1 + l_n l_a e^{-2/\tau}) \xrightarrow{\tau \to 0^+} 0$. □ □

E. Pathology 3: Heuristic Loss Weighting

The objective $\mathcal{L}^{en} + \mathcal{L}^{de}$ uses equal weights with no justification. Kendall et al. [9] show the Pareto-optimal weighting depends on per-task homoscedastic uncertainty and cannot be set universally. Equal weighting is particularly unjustified in the gradient-conflict regime where the encoder wants discriminative compression and the decoder wants reconstruction fidelity.

IV. GS-CRC: The Proposed Loss

A. Adaptive Temperature

Theorem 4 (Optimal batch temperature)**.** *Let negative-pair cosine similarities concentrate with standard deviation σ_h and $l_a^{(\min)} = \max(1, \lfloor \delta l_a \rfloor)$. The temperature minimising worst-case per-subclass gradient suppression is*

$$\tau^* = \frac{\sigma_h}{1 + \log\left(l_n l_a / l_a^{(\min)}\right)} \cdot \sqrt{2 \log(l_n l_a)}. \quad (6)$$

For UNSW-NB15 ($\sigma_h \approx 0.12$, $\mathrm{bs} = 1024$): $\tau^* \approx 0.08$, roughly $4\times$ larger than the paper's $\tau = 0.02$.

Proof sketch. The log-gradient on subclass c is $-\log \tau + \bar{h}^{(c)}/\tau - \log \sum_{c'} l_a^{(c')} e^{\bar{h}^{(c')}/\tau}$. Maximise worst-case via KKT on the log-sum-exp envelope; interior optimum solves $\tau^2 \partial_\tau \mathrm{LSE} = \tau$, yielding (6). □ □

In practice $\hat{\sigma}_h(t)$ is estimated per batch from the inter-class cosine similarity matrix and $\tau_0(t) = \tau^*(\hat{\sigma}_h(t))$ is clamped to $[0.01, 0.50]$.

B. Definition of GS-CRC

Definition 1 (GS-CRC loss). *For batch $\mathcal{B} = \mathcal{B}_n \cup \mathcal{B}_a$ with unit-normalised embeddings $\{z_i\}$, define positive set $\mathcal{P}_i = \{j \neq i : y_j = y_i\}$ and negative set $\mathcal{N}_i = \{k : y_k \neq y_i\}$. Per-anchor loss:*

$$\mathcal{L}_i^{\text{GS-CRC}} = -\log \frac{\sum_{j \in \mathcal{P}_i} e^{h(i,j)/\tau_{ij}}}{\sum_{j \in \mathcal{P}_i} e^{h(i,j)/\tau_{ij}} + \rho^{(y_i)} \sum_{k \in \mathcal{N}_i} w_{ik}^- \, e^{h(i,k)/\tau_{ik}}}, \tag{7}$$

with components:

1) Adaptive temperature*: $\tau_{ij} = \tau_0(t)(1 + \beta|h(i,j) - \bar{h}_{y_i}|)$, $\tau_0(t)$ from* (6).
2) Focal negative weights*: $w_{ik}^- = (1 - s_{ik})^\gamma$, $s_{ik} = e^{h(i,k)/\tau_0} / \sum_{k' \in \mathcal{N}_i} e^{h(i,k')/\tau_0}$.*
3) Class-balance prefactor*: $\rho^{(n)} = l_n/l_a$, $\rho^{(a)} = l_a/l_n$.*
4) Bounded influence*: clip each $\mathcal{L}_i$ at $\mathcal{L}_{\max}$.*

Total loss: $\mathcal{L}^{\text{GS-CRC}} = \frac{1}{l_n}\sum_{i \in \mathcal{B}_n} \mathcal{L}_i + \frac{\alpha_{symm}}{l_a}\sum_{i \in \mathcal{B}_a} \mathcal{L}_i$. Setting $\gamma = 0$, $\alpha_{symm} = 0$, $\rho = 1$, $\mathcal{L}_{\max} = \infty$, $\tau_{ij} = \tau$ recovers CRC exactly.

C. Theoretical Properties

Theorem 5 (Fisher consistency of GS-CRC). *Under any joint distribution P with $\pi_a > 0$ and finite KL divergence between $p(\cdot|y = 0)$ and $p(\cdot|y = 1)$, the population minimiser of $\mathcal{L}^{\text{GS-CRC}}$ over a sufficiently expressive encoder yields a representation on which the Bayes classifier is realisable as a linear separator.*

Proof sketch. The class-balance prefactor $\rho^{(y_i)}$ makes empirical sums unbiased estimators of population expectations. The symmetric structure decomposes the population risk into four terms: $\mathcal{L}_n^+$ (within-normal cohesion), $\mathcal{L}_n^-$ (normal-from-abnormal repulsion), $\mathcal{L}_a^+$ (within-abnormal cohesion, absent in CRC), and $\mathcal{L}_a^-$ (abnormal-from-normal repulsion). The term $\mathcal{L}_a^+$ prevents abnormal collapse to a single antipodal point: any abnormal-class structure reducing within-class spread is preferred. Under the reconstruction constraint, this forces a linearly Gaussian-separable geometry. □ □

Theorem 6 (Polynomial gradient balance). *Under GS-CRC with $\gamma \geq 1$ and adaptive τ_{ij}, the per-sample gradient norm on subclass c satisfies:*

$$\left\|\nabla_{z_m} \mathcal{L}^{\text{GS-CRC}}\right\| \;\geq\; \frac{C}{\tau_0 \, l_a} \cdot \left(1 - s_{\max}^{(c)}\right)^\gamma. \tag{8}$$

For $\gamma = 2$ and near-uniform softmax ($s_{\max}^{(c)} \lesssim 1/l_a$), the bound is $\Omega(1/l_a)$. There is no exponential term in Δ/τ.

Proof. The focal weight $(1 - s_{ik})^\gamma$ bounds the dominant-subclass contribution uniformly: $s(1 - s)^\gamma \leq 1/(\gamma + 1)$ even as $s \to 1$. The effective denominator $S^* \leq S(1 - s_{\max})^\gamma$, so $\bar{Q}^* \geq 1/S$. The exponential $e^{(\bar{h}^{(c)} - h^{(\max)})/\tau}$ from Theorem 2 is absorbed by the focal weight. □ □

The contrast is the central result: CRC imposes exponential starvation; GS-CRC imposes polynomial decay.

D. Uncertainty-Weighted Multitask Loss

Following Kendall et al. [9]:

$$\mathcal{L}_{\text{final}} = \frac{\mathcal{L}^{\text{GS-CRC},en}}{2\sigma_e^2} + \frac{\mathcal{L}^{\text{GS-CRC},de}}{2\sigma_d^2} + \log(\sigma_e \sigma_d), \tag{9}$$

where $\sigma_e, \sigma_d > 0$ are trained jointly with encoder and decoder.

V. Decision-Making Improvements

A. GMM Excess Risk

AOC-IDS fits a 2-component Gaussian to the full abnormal cosine-similarity score distribution. UNSW-NB15 has 10 attack categories; their cosine similarity scores to the normal centroid form a K-modal distribution, not a single Gaussian.

Theorem 7 (GMM excess risk). *Suppose abnormal scores follow a K-component mixture with means $\mu_{a,1} < \cdots < \mu_{a,K}$, weights w_k, and within-component variances $\sigma_{a,k}^2$. Fitting a single Gaussian gives merged parameters $\bar{\mu}_a = \sum_k w_k \mu_{a,k}$ and $\bar{\sigma}_a^2 = \sum_k w_k(\sigma_{a,k}^2 + (\mu_{a,k} - \bar{\mu}_a)^2)$. The excess Bayes risk over the K-component oracle satisfies*

$$\Delta\mathcal{R} \geq \frac{1}{2} \sum_k w_k \, \frac{(\mu_{a,k} - \bar{\mu}_a)^2}{\bar{\sigma}_a^2} \cdot (1 - p_{\text{far}}^{(k)}), \tag{10}$$

where $p_{\text{far}}^{(k)}$ is the probability that a class-k sample falls outside the merged model's decision boundary. This excess is $\Omega(1)$—it does not vanish with sample size.

Proof sketch. The oracle boundary for component k lies at the crossing of $p(s|a, k)$ and $p(s|n)$, while the merged model uses $p(s|\bar{a})$ vs $p(s|n)$. The probability mass misclassified due to this shift is lower-bounded by w_k times the squared standardised displacement from the merged mean, corrected by the non-overlap fraction. Since the between-component variance does not shrink with N, the excess is $\Omega(1)$. □ □

Fix. Use a K-component GMM with K selected by BIC up to $K_{\max} = 10$ for UNSW-NB15 (one component per attack category). The abnormal log-likelihood is then $\log \sum_k w_k \, p(s|\mu_{a,k}, \sigma_{a,k})$ — the same fitted densities used in the LLR, with no additional computation.

B. LLR Fusion

Proposition 8 (LLR decision rule). *Under conditional independence of encoder and decoder scores s_e, s_d given y, the Bayes-optimal prediction is*

$$\hat{y} = \begin{cases} 1 & \text{if } \Lambda_e + \Lambda_d + \log(\pi_n/\pi_a) < 0, \\ 0 & \text{otherwise}, \end{cases} \tag{11}$$

where $\Lambda_e = \log p(s_e|y = 0) - \log p(s_e|y = 1)$ and Λ_d analogously, with priors π_n, π_a estimated from the current pseudo-label distribution.

In the K-GMM setting, $\log p(s_e|y{=}1)$ is the K-component log-likelihood from the fitted mixture; the decision rule is otherwise identical. The prior term $\log(\pi_n/\pi_a)$ provides a calibrated operating-point knob: increasing it shifts toward higher precision; decreasing it toward higher recall.

C. Information-Theoretically Optimal Bottleneck

Proposition 9 (Optimal bottleneck dimension)**.** *Under the Information Bottleneck principle [10] with K subclasses and intrinsic data dimensionality d^*, the bottleneck minimising $I(X;Z) - \beta I(Z;Y)$ is*

$$m^* = \lceil d^* + \log_2 K \rceil. \tag{12}$$

For UNSW-NB15 ($d^* \approx 28$, $K = 10$): $m^* = 32$, vs the paper's $m = 64$. The over-parameterised bottleneck facilitates the degenerate encoder family of Theorem 3.

VI. Systems-Level Improvements

A. Drift-Adaptive Update Cadence

AOC-IDS retrains the model every fixed 2,000 samples regardless of whether the distribution has actually shifted.

Proposal. Maintain the current GMM G_t. After each incoming batch, fit a candidate G_t^* and compute the symmetric KL divergence $D_{\text{drift}}(t) = \frac{1}{2}[\text{KL}(G_t \| G_t^*) + \text{KL}(G_t^* \| G_t)]$. Trigger model retraining only if $D_{\text{drift}}(t) > \epsilon$.

Theorem 10 (Optimal drift threshold)**.** *Under risk tolerance η (maximum acceptable F1 degradation between updates) and Lipschitz constant L relating distribution drift to risk:*

$$\epsilon^* = \frac{\eta^2}{8L^2}. \tag{13}$$

Proof. By Pinsker's inequality, $\|G_t - G_t^*\|_{\text{TV}} \le \sqrt{D_{\text{drift}}/2}$. The risk gap satisfies $|\mathcal{R}(G_t) - \mathcal{R}(G_t^*)| \le L\sqrt{D_{\text{drift}}/2} \le \eta$, giving $D_{\text{drift}} \le 2\eta^2/L^2$. The factor-of-4 margin in (13) provides conservative headroom. □ □

For $\eta = 0.01$, $L \approx 1$: $\epsilon^* \approx 1.25 \times 10^{-5}$. In the UNSW-NB15 simulation, adding 2,000 pseudo-labeled samples each round produces KL drift consistently above this threshold (observed range: 10^{-2} to 10^{+1}), so the trigger does not skip retrains in this setting. The benefit manifests in real deployment under stable traffic, where the trigger avoids unnecessary retraining.

B. EMA Normal Centroid

AOC-IDS fixes the normal centroid to the initial training set, preventing adaptation to benign traffic drift.

Proposal. At each online round t, update: $\hat{\mu}_n^{(t+1)} = (1 - \alpha_t)\hat{\mu}_n^{(t)} + \alpha_t \bar{z}_{\text{pred-n}}^{(t)}$, where $\bar{z}_{\text{pred-n}}^{(t)}$ is the mean embedding of samples predicted as normal in the current batch, renormalised to the unit sphere.

Theorem 11 (EMA drift-tracking bound)**.** *If the true normal mean drifts at speed $\|\mu_n(t+1) - \mu_n(t)\| \le v$ and per-batch noise is $\sigma_z/\sqrt{B}$, then $\alpha^* = \min\{1, \sqrt{vB/\sigma_z}\}$ gives*

$$\mathbb{E}\|\hat{\mu}_n^{(t)} - \mu_n(t)\|^2 \le \frac{2\sigma_z v}{\sqrt{B}} + O(v^2), \tag{14}$$

which is order-optimal among constant-step EMA estimators.

Proof sketch. MSE under step α decomposes as $\alpha^2\sigma_z^2/B$ (noise) $+v^2/(2\alpha^2)$ (lag). Minimising over α gives $\alpha^* = (vB/\sigma_z)^{1/2}$ and the stated bound. □ □

TABLE I
EXPERIMENTAL CONFIGURATION

Parameter	CRC (Baseline)	GS-CRC (Ours)
Loss	CRC	GS-CRC (Def. 1)
Bottleneck m	64	32 (Prop. 9)
Decision rule	Majority vote	LLR (Prop. 8)
Abnormal discriminator	2-Gaussian	K-GMM, BIC $K \le 10$
Normal centroid	Fixed	EMA $\alpha = 0.05$ (Thm. 11)
Update trigger	Fixed interval	KL-drift $\epsilon^* = 10^{-5}$
Loss weighting	Equal	Learned σ_e, σ_d
Batch size	256	1024
Epochs	100	100
γ	–	2.0
α_{symm}	–	0.5
β_{temp}	–	0.5
$\mathcal{L}_{\max}$	–	5.0
Optimizer	SGD, 10^{-3}	SGD, 10^{-3}
PyTorch	1.13.1	2.11.0+cu130
GPU	RTX 5060 Laptop	RTX 5060 Laptop

TABLE II
PER-SEED RESULTS ON UNSW-NB15 (100 EPOCHS)

	Seed	Acc	Prec	Rec	F1
GS-CRC (ours)	5009	0.8456	0.8092	0.9417	0.8704
	5010	0.8409	0.7972	0.9536	0.8684
	5011	0.8379	0.7852	0.9714	0.8684
	5012	0.8320	0.7878	0.9510	0.8617
	5013	0.8430	0.8015	0.9502	0.8695
	Mean	**0.8399**	**0.7962**	**0.9536**	**0.8677**
	Std	0.0049	0.0087	0.0107	0.0032
CRC (baseline)	5009	0.5506	0.5506	1.0000	0.7102†
	5010	0.4494	0.0000	0.0000	0.0000*
	5011	0.4827	1.0000	0.0605	0.1141*
	5012	0.5697	0.5749	0.8387	0.6822
	5013	0.4494	0.0000	0.0000	0.0000*
	Mean	0.5004	0.4251	0.3798	0.3013
	Std	0.0537	0.3905	0.4507	0.3151

† Degenerate: all-attack. * Collapsed: all-normal.

VII. Experiments

A. Setup

We evaluate on UNSW-NB15 [11] (196 features, MinMax normalised). All hyperparameters other than the loss, architecture, and pipeline match AOC-IDS exactly to isolate the contribution. The run at 100 epochs is one-eighth of the paper's standard 800.

Test set: 37,000 normal (44.9%) and 45,332 attack (55.1%). Trivial all-attack baseline: accuracy 55.1%, F1 0.710. Trivial all-normal baseline: accuracy 44.9%, F1 0.000.

B. Per-Seed Results

C. Comparison Summary

D. Confusion Matrices

E. Analysis

Stability. All five GS-CRC seeds produce non-degenerate confusion matrices (0/5 collapses, down from 3/5). Theorems 3 and 5 predict exactly this: Fisher inconsistency allows CRC to

TABLE III
SUMMARY COMPARISON (MEAN ± STD, 5 SEEDS, 100 EPOCHS)

Method	Accuracy	Precision	Recall	F1
CRC (base)	0.500±.054	0.425±.391	0.380±.451	0.301±.315
GS-CRC	**0.840**±.005	**0.796**±.009	**0.954**±.011	**0.868**±.003
Δ (abs.)	+0.340	+0.371	+0.574	+0.567
Δ (%)	+68.0%	+87.3%	+151.1%	+188.2%

TABLE IV
CONFUSION MATRICES, SELECTED SEEDS

	Pred Normal		Pred Attack	
Method	TN	FP	FN	TP
CRC seed 5009	0	37000	0	45332
GS-CRC seed 5009	26933	10067	2644	42688
CRC seed 5010	37000	0	45332	0
GS-CRC seed 5010	26003	10997	2103	43229
CRC seed 5012	8884	28116	7312	38020
GS-CRC seed 5012	25391	11609	2223	43109

settle into degenerate geometry with positive probability, while GS-CRC's $\mathcal{L}_a^+$ term rules it out. F1 std drops from 0.315 to 0.003 — a 99-fold stability improvement.

Recall improvement. Mean recall rises from 0.380 to 0.954 (+151.1%). The K-component GMM correctly models the multi-modal abnormal score distribution (10 attack categories produce 10 distinct cosine-similarity clusters), recovering the $\Omega(1)$ excess risk identified in Theorem 7. The GS-CRC loss additionally ensures that minority-subclass representations receive polynomial rather than exponential gradient, so all 10 attack subtypes remain separated in the embedding space by round 100.

Precision–recall tradeoff. Precision settles at 0.796 ± 0.009, down from 0.998 ± 0.000 in the 2-Gaussian version. The K-GMM models the abnormal distribution with more components, raising $\log p(s|a)$ in overlap regions between low-frequency attack subtypes and the normal class. This shifts the LLR threshold, increasing sensitivity (recall) at the cost of specificity (false positive rate $\approx 28\%$). The false positive rate is controllable without retraining: increasing $\log(\pi_n/\pi_a)$ in (11) shifts the operating point toward higher precision along the precision–recall curve.

Baseline artefact. CRC seed 5009 achieves F1 $= 0.710$ by predicting every sample as attack, matching the trivial all-attack classifier exactly (recall $= 1$, precision $= 0.551$). This is not a detection result.

VIII. LIMITATIONS AND FUTURE WORK

Epoch budget. All results are at 100 epochs (one-eighth of the standard 800). The 800-epoch run on both CRC and GS-CRC is the required next experiment for definitive numbers.

Dataset scope. We evaluate only on UNSW-NB15. NSL-KDD evaluation is needed to confirm that gradient-balance improvement transfers to $K = 5$ attack types and a different imbalance structure ($l_a^{(\min)}/l_a \approx 0.001$ for U2R).

Parameter sweeps. γ, α_{symm}, β_{temp}, and $\mathcal{L}_{\max}$ are set by theoretical argument rather than grid search. A full ablation is left to the extended version.

KL-drift trigger calibration. In the UNSW-NB15 simulation, adding 2,000 samples per round produces KL drift consistently above $\epsilon^* = 10^{-5}$, so the trigger retrains every round. For deployment under stable traffic the trigger would skip unnecessary retrains; calibrating η and L for a specific network environment is future work.

IX. CONCLUSION

We presented a comprehensive upgrade to AOC-IDS addressing seven structural pathologies. The GS-CRC loss fixes five defects in the CRC objective via symmetric structure, focal modulation, adaptive temperature, class-balance prefactors, and bounded influence. The detection pipeline is upgraded with a K-component GMM discriminator (fixing $\Omega(1)$ excess risk), an EMA normal centroid (fixing static drift), a KL-drift update trigger (fixing arbitrary cadence), and a calibrated LLR decision rule.

On UNSW-NB15 at 100 epochs, the full system raises mean F1 from 0.301 to 0.868 (+188.2%), recall from 0.380 to 0.954, eliminates all five degenerate seeds, and reduces F1 standard deviation from 0.315 to 0.003. Each improvement is backed by a formal theorem rather than an empirical observation, making the contribution robust to dataset and deployment variation.

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